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PCM-Filled Capsules (RT44HC) for Heat Storage—Laboratory Scale Pilot Study

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Abstract: Peak power shaving in heating systems can be achieved using heat accumulators, traditionally implemented in the form of water storage tanks. Their heat capacity can be increased by using a phase change material (PCM) instead of water, which, however, usually requires a change in the tank design. The innovation of this paper is an interesting concept to use plastic capsules filled with a PCM that replace part of the water volume in an existing heat accumulator. The aim of this paper is to compare the cooling rate of the same volume of water as that of the water mixed with the PCM capsules to initially verify the heat storage potential of the capsules. The results of pilot experimental studies on a laboratory scale are presented and discussed, showing the potential of this idea for heat storage. The partial replacement of water with capsules (40% of the total volume) results in significantly faster heat accumulation with the same tank volume (3.85 times at the beginning of the process) and more heat stored (decrease in the temperature of water alone by 14 K and water with PCM capsules by 26 K in the same period of time), which gives promising perspectives for the use of this solution on a semitechnical scale and further in a real-size heating system.

Keywords: PCM; heat storage; RT44HC; district heating; heat accumulation

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1. Introduction

Energy storage is gaining importance in the context of the use of renewable energy sources, the availability of which usually is not synchronized with the energy needs of society. Currently, due to the growing share of wind farms and photovoltaic (PV) panels in energy systems, great emphasis is placed on research in the field of electrical energy storage [1]. However, the topic of thermal energy storage is also important from the point of view of modern energy supply systems. On the way to carbon neutrality, existing heating systems are undergoing successive modernizations, as we wrote earlier in the article [2]. It turns out that one of the transitional steps in this process is the use of gas cogeneration (CHP), which allows for an increase in the energy efficiency of the electricity and heat generation process and also the possibility of delivering additional financial profit for the energy company. Cogeneration engines require the need to ensure an appropriate operating temperature. In addition, there is often a situation when it is profitable to produce more electricity due to the favorable price ratio of gas and electricity, which causes the generation of excess waste heat, and therefore, the need for its storage is crucial and needed. In [3], we have shown that storing heat in a distributed heating system leads to an improvement in its energy efficiency. This is not the only case where heat storage

seems to be very desirable. In heating systems powered by a heat pump, it is typical: an additional tank improves the operating conditions of the device itself, which is switched on less often and can operate for longer with high efficiency. On the other hand, in the international literature, there are works that analyze the cooperation of a heat pump with a heat storage tank to shave the peak power demand for electricity, for example, [4]. It turns out that decarbonization through the electrification of heating systems using heat pumps is associated with a significant load on the existing electricity distribution system, which can be relieved by using appropriate strategies to load and unload heat storage tanks [5]. In turn, in [6], we described the subject of difficulties in selecting the appropriate thermal power of the heat source or the heat substations for the preparation of domestic hot water (DHW). The stochastic nature of hot water collection encourages the use of a buffer tank, which can reduce the peak power, reducing the peak demand for thermal power [7], but also improve the comfort of hot water use (more stable supply) and ultimately reduce the demand for non-renewable primary energy (PE) [8] and the consumption of fossil fuels [9]. In turn, high thermal comfort for users of modern buildings with low energy demand [10] can be achieved by using low-temperature heating systems [11], including local ones [12], which support the use of renewable energy sources (RES). Thanks to heat storage, such buildings can be supplied with RES, achieving a higher level of thermal comfort among users [13]. Similarly, waste heat storage may be advisable in the case of recently increasingly popular heat recovery systems, e.g., from exhaust gases from gas-type heaters [14], from exhaust ventilation air [15–17], or even from domestic sewage [18–20], which was broadly analyzed in the review article [21].

An interesting possibility of improving the efficiency of heat storage is provided by PCMs. In article [22], we describe the case of using dispersed PCM heat accumulators in a district heating system, which allowed for system efficiency to increase by 41%. Additionally, the LCA (Life Cycle Analysis) showed that a positive impact on the environment was achieved, expressed as negative values of the Eco-indicator from -0.504×10^{-2} to -6.44×10^{-2} kPt/kW. In the other case [23] of LCA, using the Eco-indicator 99 (EI99) method, PCMs RT-27 paraffin and SP-25 salt hydrate were compared in order to have a positive environmental impact of around 10%. The available literature contains research results that mainly focus on the thermochemical properties of a PCM. For example, in the article [24], the properties of paraffin with a melting point of 58 °C were experimentally determined, analyzing its suitability for energy storage in waste heat recovery systems and the preparation of domestic hot water (DHW). The article determined the "thermal and cycling stability, health hazard, phase change thermal range, enthalpy and specific heat". In the article [25], the thermal characteristics of a bio-PCM in the form of beeswax were experimentally determined during the charging and discharging of the heat accumulator. The charging duration was also analyzed. In the article, a less typical compound was analyzed: magnesium hydroxide, $Mg(OH)_2$ [26]. Molecular dynamics simulations showed that enriching magnesium hydroxide with aluminum oxide (Al_2O_3) nanoparticles allows for 12.9% higher thermal conductivity of a PCM, which is supposed to facilitate more efficient energy storage. In another work [27], nano-SiO_2 and nano-MgO were dosed with different salts, $LiNO_3$, $NaNO_3$, KNO_3, $NaNO_2$, and KNO_2, resulting in a maximum 50.4% increase in specific heat and 19.2% increase in the thermal conductivity of the base salt. In the article [28], the PCM composite of octanoic acid–tetradecanol was mixed with modified expanded graphite, obtaining a 133% increase in thermal conductivity. The overview of the thermal properties of other different organic PCMs can be found in the review paper [29]. In the above works, the research was conducted on a laboratory scale, using a PCM directly. In our work, we focused on a rare method of using a PCM enclosed in capsules.

The use of a PCM for heat storage usually requires a different type of heat storage tank from that used for heat storage in water. This is reported by, for example, the authors of the article [30], who optimized the tank design using ribs to improve the intensity of heat exchange. There are many examples of the use of shell-tube heat exchangers for heat storage with the participation of a PCM in the literature. For example, in the article [31], this type of heat exchanger was used for heat storage in PCM-based nanofluids, focusing on improving the energy storage parameters in relation to water systems. The energy storage and release performance of the shell-tube heat exchanger filled with a PCM in the form of paraffin was the subject of research presented in [32]. The article focused on changing the geometric parameters of the tank and its effect on the heat storage efficiency. In turn, in the paper [33], the "effects of PCM inlet pressure, port geometry, and positioning" were investigated using artificial intelligence algorithms. In this way, the influence of these design parameters on the melting behavior of a PCM in the exchanger was presented.

These examples show that heat accumulators are not very suitable for PCMs to be filled with to improve their thermal capacity. Cost- and time-consuming constructional changes are needed, while the operation of the system has to be stopped. An interesting intermediate solution may be the use of PCM-filled plastic capsules, which in existing tanks would replace part of the volume of water from the tank. This is a solution that does not require changing the tank design or using additional elements to intensify heat exchange. Compared to water, PCM accumulators require about 40% smaller tank volume [34]. In this situation, the risk in tank destruction related to PCM thermal expansion is also eliminated because the PCM is in its own housing, which limits volume changes. In this article, we present the results of pilot studies on a laboratory scale with the aim of analyzing the potential of using capsules filled with a PCM in existing water energy accumulators. Recently, one can read about a similar idea of using a PCM in capsules, e.g., in the article [35]. In this case, a PCM in the form of paraffin was enclosed in 60 mm spherical capsules made of 1 mm stainless steel. The capsules were used to build a packed bed thermal energy storage tank, and then, the geometric parameters of the capsule system and their effect on the thermal efficiency of the storage tank were analyzed. In another work [36], the effect of uneven flow distribution was investigated in a PCM bed-type heat storage tank. The results show that "neglecting variable porosity leads to underprediction of the bed's charging time". Capsules of a diameter of 55 mm filled with paraffin with a specific heat of 1850 J/(kgK) were used. The capsule material and wall thickness were not presented. In our research, we used a different type of capsules—cuboid shaped, made of plastic, and usually used in portable tourist refrigerators as cold storage inserts. These capsules are filled with RT44HC salt. This material was used in other studies, e.g., in [37], where the cooperation of a thermal storage tank filled with capsules with RT44HC salt of 20 mm diameter was analyzed. However, the material of the capsule housing was not given. The capsule was a sphere, unlike our capsules that were in the shape of a cuboid. The authors of the work demonstrated the advantages of using a heat storage tank for a PV panel system that cooperates with a heat pump as an exemplary application.

Taking into account the above, an approach such as the one presented in this article or similar studies have not been found in the available literature. Plastic capsules of cuboid shape filled with the PCM RT44HC have not been previously studied in the context of their technical use in heat storage systems, in contrast to numerous publications on the phase change material itself or on the mentioned modifications of tank structures while using them for storage of a PCM instead of water. In turn, the results of the current studies indicate an increase in the heat removal rate by a PCM compared to water, giving promising prospects for the use of this solution on a semitechnical scale and in real full-scale heating systems.

2. Materials and Methods

To verify the possibility of using PCM RT44HC filled plastic capsules for heat storage purposes in existing water tanks, experimental studies were conducted. The experiment consisted of the following:

- Measuring the water temperature in a tank filled with water during its cooling from a temperature of approx. 70 °C (at a frequency of every 10 s)—case 1;
- Measuring the water temperature in a tank filled with water and PCM capsules during its cooling from a temperature of approx. 70 °C (at a frequency of every 10 s)—case 2;
- Comparing the results in order to detect the difference in cooling time, which would suggest the appearance of the accumulation effect due to the shortened heat collection time.

The tests used a tank with a capacity of 9.6 dm^3 and external dimensions: 32.5 × 38.5 × 18.5 cm. The schema of the tank with its dimensions is presented in Figure 1. A photo of the tank is shown in Figure 2. The tank was insulated on each side with Styrofoam with a thickness of g = 0.02 m and a thermal conductivity coefficient of λ = 0.032 W/(mK), constituting an additional thermal resistance (R) equal to the value of

$$R = \frac{g}{\lambda} = \frac{0.02}{0.032} = 0.625 \left[\frac{m^2K}{W}\right].$$

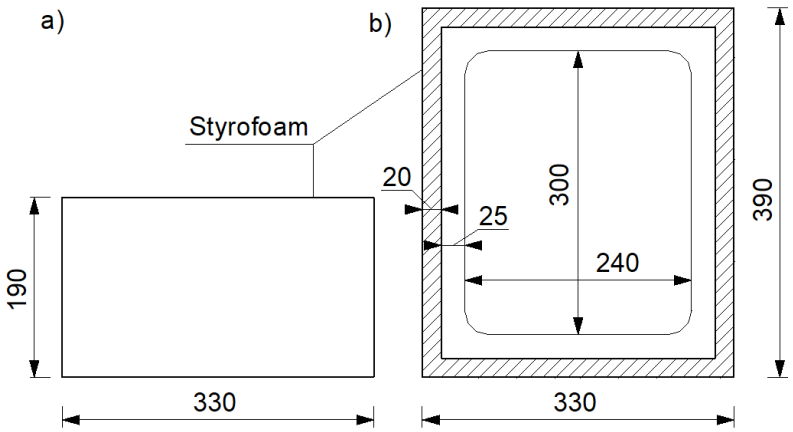


Figure 1. Dimensions in [mm] of the tank used during the tests: (**a**) side view and (**b**) top view.



Figure 2. A photo of the tank used during the tests: visible Styrofoam insulation.

Nine PCM-filled capsules were placed in the tank. A photo of the single capsule and dimensions is shown in Figure 3.

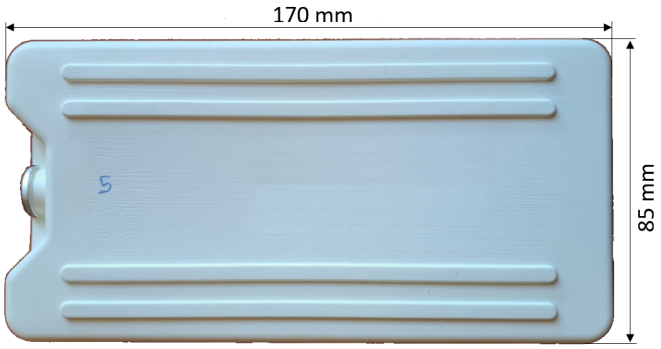


Figure 3. A photo of a capsule filled with the PCM used during the investigation (thickness 22 mm).

The volume of the capsules was determined by immersing them in a laboratory cylinder with a volume scale with a 10 mL division. The increase in water volume after the capsule was completely immersed in the cylinder was measured. All capsules used in the study were weighed using a laboratory scale. The results of these measurements are presented in Table 1.

Table 1. Results of the measurements of the weight and volume of the capsule filled with the PCM.

Capsule	Mass (g)	Volume (mL)
1	236	300
2	241	305
3	243	300
4	245	295
5	238	300
6	239	295
7	243	300
8	237	305
9	238	300
Average	240	300

One of the capsules (no. 5) was cut open (see Figure 4), and using the same methods, the mass and volume of a single capsule were determined:

- Capsule mass: 238 g;
- Capsule volume: 295 mL;
- Capsule casing mass (plastic): 38 g;
- Capsule casing volume (plastic): 70 mL;
- PCM mass in the capsule: 200 g.



Figure 4. A photo of the interior of the capsule—the PCM RT44HC in the solid phase.

The PCM parameters contained in the capsules are presented in Table 2.

Table 2. Properties of the PCM used in the experiment [38].

Properties	Value
Melting area	41–44 °C
Congealing area	44–40 °C
Main peak	43 °C
Specific heat, c_p	2.0 kJ/(kgK)
Density, ρ (solid, at 25 °C)	800 kg/m^3
Density, ρ (liquid, at 80 °C)	700 kg/m^3
Thermal conductivity in both phases, λ_{PCM}	0.2 W/(mK)
Volume expansion	12.5%

The list of measuring devices used during all measurements in the experiment is presented in Table 3.

Table 3. The measurement equipment with the accuracy used in the experiment.

Description	Accuracy
Testo 176 T2 data logger (Celsiusstr. 2, Titisee-Neustadt, 79822, Germany), resolution: 0.01 °C, PT100 A-class sensors, overall temperature measurements accuracy (given by manufacturer)	±0.2 K
Laboratory scale	±1 g
Laboratory cylinder with a volume scale till 1000 mL	±5 mL
Liquid thermometer for measuring ar temperature in the laboratory	±1 K

Regarding the analysis of measurement uncertainty, in the studies carried out, the dominant component of uncertainty was the systematic component because the studies consisted of measuring the values of physical quantities obtained directly. In such a situation, the value of the smallest scale division of the measuring instrument or a fraction of the scale division, determined by the class of the instrument, is called the reading accuracy and is the maximum value of the systematic uncertainty introduced by the measuring instrument itself. The maximum uncertainty of the partial measurements did not exceed the values presented in Table 4. For repeated/multiple measurements (e.g., reading the volume indication twice—at the beginning of the water only and after inserting the capsule), the error was calculated according to Equation (1).

$$\Delta x = \sqrt{\sum_{i=1}^{n} \Delta x_i^2} \quad (1)$$

where

Δx—multiple measurement uncertainty;

Δx_i—uncertainty of single measurement.

Table 4. Measurement uncertainties.

Description of Uncertainty	Symbol	Measurement Uncertainty
Temperature of water (single measurement)	ΔT_1	2×10^{-1} °C
Temperature of air (single measurement)	ΔT_{air}	1 °C
Temperature of water (as an average from four readings)	ΔT_{av}	4×10^{-1} °C
Mass (single measurement)	Δm	10^{-3} kg
Volume (single measurement)	ΔV_1	5×10^{-3} dm^3
Volume of capsule (two readings)	ΔV	7.1×10^{-3} dm^3

3. Results

3.1. Quantitative Research

Using the procedure described in point 2, experimental studies were carried out to quantitatively compare the cooling process of water in a tank filled with water only (case 1) or filled with water with PCM capsules (case 2—see Figure 5). For the purposes of conducting the experiment for case 1, hot water with a volume of 6750 mL and a temperature of approx. 75 °C was poured into the insulated tank, and then, the water temperature was measured at two points from the opposite sites of the tank. For the purposes of conducting experiment case 2, hot water in the amount of 4050 mL was poured into the insulated tank and nine capsules filled with the PCM (see Figure 5) with an average volume of 300 mL for each (total volume of capsules = 2700 mL) were placed, obtaining the same value of a total volume of 6750 mL. The capacity of the capsules that replaced the same volume of water was 2700 mL = 40% of the total volume of the water and PCM mixture. The water temperature was selected so that after a stabilization period of about 1 min, the temperature of the water and PCM mixture was about 70 °C. Then, the water temperature was measured at two points as in case 1. The average air temperature in the laboratory during the measurements was 27 °C ± 1 °C. Both variants of the tests, case 1 and case 2, were carried out under the same initial conditions in terms of the initial temperature of the tank itself in order to eliminate the influence of the heat capacity of the tank material on the results of the comparison. The results of the change in water temperature over time, calculated as the average of the values measured by two temperature sensors, are presented in Figures 6 and 7. The results were averaged using measurements carried out in two measurement series (two series for case 1 and two series for case 2).



Figure 5. The top view of the tank filled with water and PCM capsules (9 pcs).

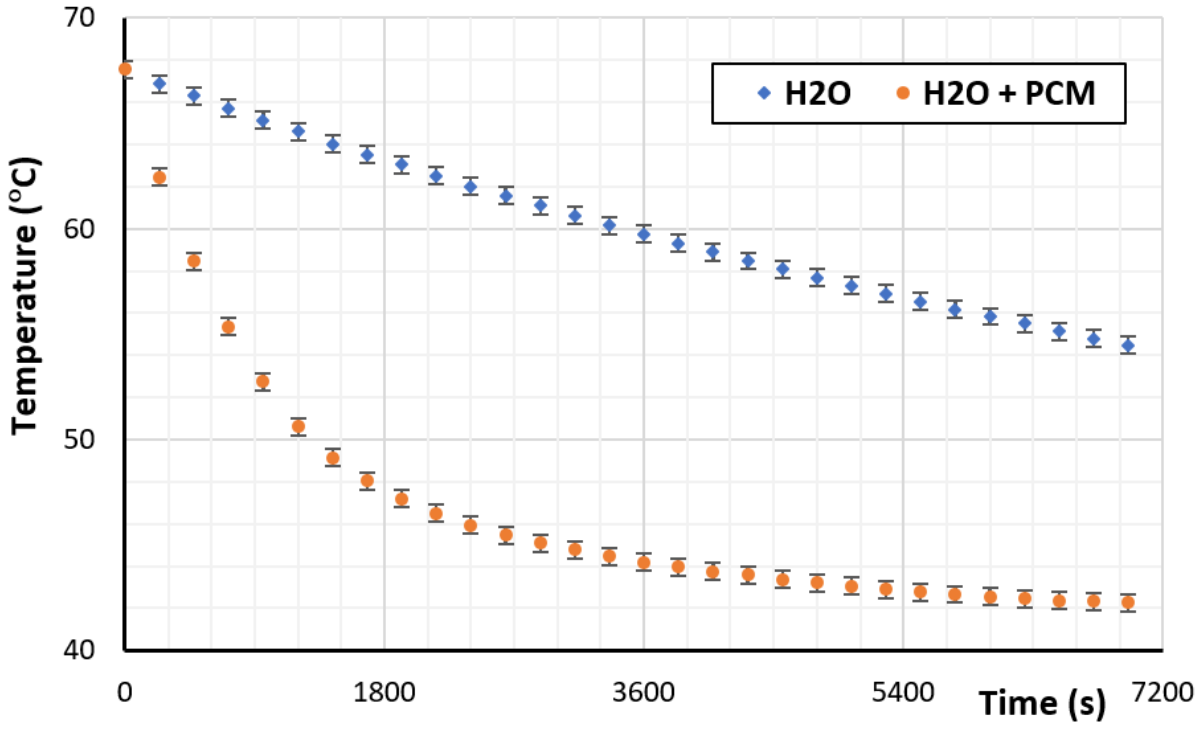


Figure 6. Averaged results of measurements with uncertainties bars for chosen measuring points.

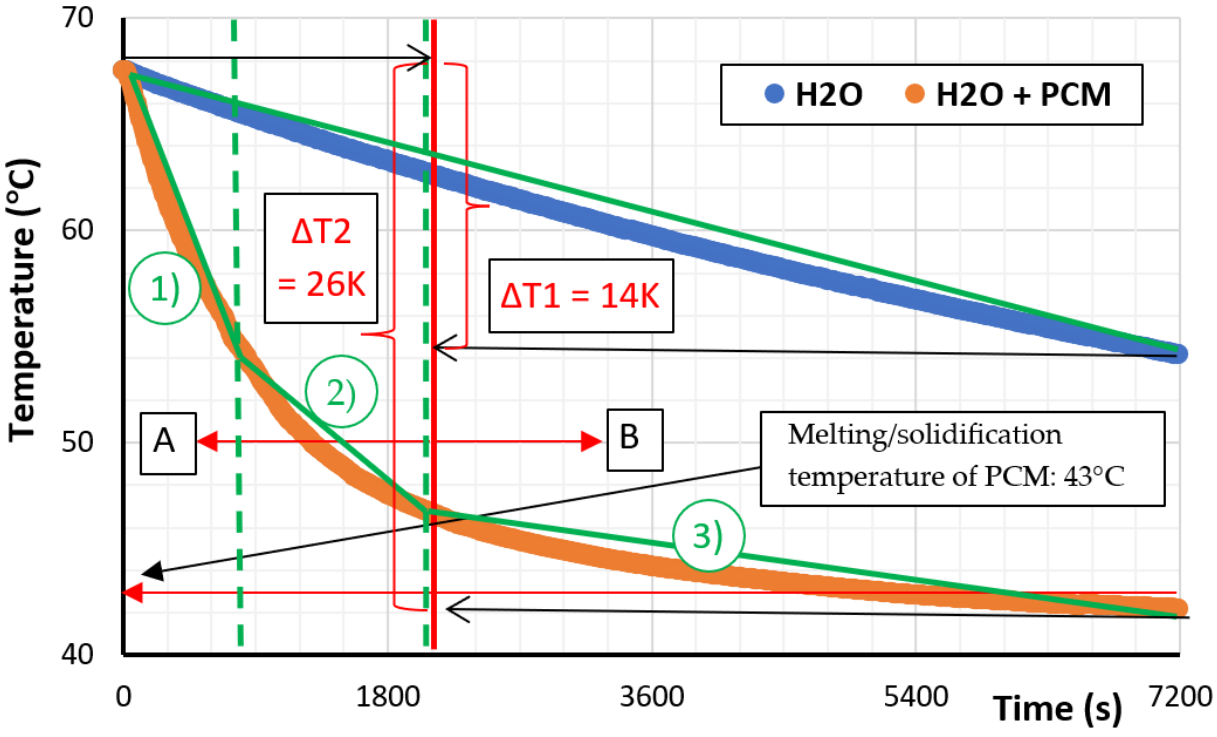


Figure 7. A comparison of the temperature of the water in a tank filled with water only (H_2O) or water with the PCM (H_2O + PCM) during cooling.

The results presented in Figure 7 show that the temperature of the water in the 2 h period dropped by ΔT1 = 14 K and of water with PCM capsules by ΔT2 = 26 K. It can also be seen that the temperature of water with capsules filled with the PCM in the first period (in Figure 7 marked A, up to approximately 2100 s) drops much faster than the temperature of water alone with the same total volume. The heat accumulated in the water volume is used in this situation to heat the capsules to the phase change temperature (approx.

43 °C) and then for the phase change purposes—the transition of the PCM from a solid to liquid state. It can be concluded from the graph that this situation occurs quite quickly, as evidenced by the rapid drop in temperature from the very beginning of the cooling process. In the range from 0 to 2100 s, the average temperature drop is 5.5/2100 = 0.0026 °C/s and 21.5/2100 = 0.010 °C/s for water and water with the PCM (water with the PCM cools 3.85 times faster).

In the second period (marked B, from approx. 2100 s to 7200 s, Δ = 5100 s), the opposite situation was observed: the temperature of water with capsules filled with the PCM drops slower than the temperature of water alone. The average rate of temperature drop is 8.5/5100 = 0.0017 °C/s and 4.5/5100 = 0.0009 °C/s for water and water with the PCM (water without the PCM cools 1.9 times faster). This is due to the smaller temperature difference between the tank contents and the environment in the case of water with the PCM (the temperature of the mixture dropped much faster than the temperature of the water). The second reason that makes the water with the PCM cool down more slowly, especially toward the end of the experiment, may be the phase change towards the solidification of the PCM, which occurs at a temperature of approx. 43 °C—i.e., close to the temperature that the temperature of the water with the PCM tends to on the graph.

In addition, the graph presented in Figure 7 shows three characteristic stages of cooling of the mixture of water and PCM capsules (marked with green lines): (1) a stage of a rapid drop in the temperature of the water in the tank (from 0 to 750 s), (2) a stage of a decreasing rate of cooling of the water in the tank (from 750 s to 2100 s, the rate of temperature drop is still greater than in the case of cooling of the water alone), and (3) a stage of a slow drop in the temperature of the water in the tank (from 2100 s to 7200 s, the rate of temperature drop is smaller than in the case of cooling of the water alone). For each of the analyzed stages, the value of the water temperature as a function of time was approximated, assuming simplified linear variability (see Figure 8). The course of the observed process was functionally described in this simplified way by functions:

$$\text{Stage (1): } t(\tau) = -0.0170\tau + 66.989,\ R^2 = 0.984,\ \Delta t/\Delta\tau = 175 \times 10^{-4}\ ^\circ\text{C/s} \quad (2)$$

$$\text{Stage (2): } t(\tau) = -0.0059\tau + 58.193,\ R^2 = 0.953,\ \Delta t/\Delta\tau = 60.8 \times 10^{-4}\ ^\circ\text{C/s} \quad (3)$$

$$\text{Stage (3): } t(\tau) = -0.0008\tau + 47.288,\ R^2 = 0.927,\ \Delta t/\Delta\tau = 9.2 \times 10^{-4}\ ^\circ\text{C/s} \quad (4)$$

$$\text{Water alone: } t(\tau) = -0.0018\tau + 66.704,\ R^2 = 0.993,\ \Delta t/\Delta\tau = 19.4 \times 10^{-4}\ ^\circ\text{C/s} \quad (5)$$

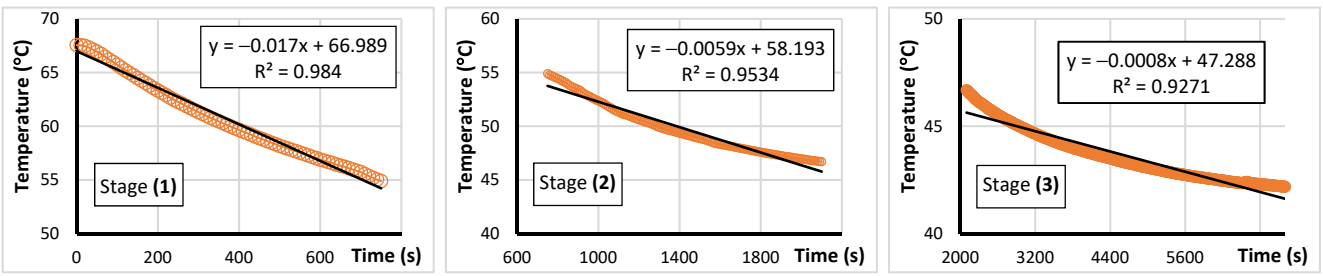


Figure 8. Three cooling stages of the water mixture with PCM-filled capsules.

3.2. Qualitative Research

As part of the research presented in this paper, an additional qualitative experiment (case 3) was conducted, during which the temperature was measured while the heat was stored in the PCM. The same proportion as in case 2 was used, i.e., 4050 mL of

water + nine capsules with the PCM with a volume of 300 mL (2700 mL = 40% of total volume) = 6750 mL. During the experiment, qualitative observations were made of changes in the consistency of the PCM substance with which the capsules were filled. The cover of the tank was opened, and all capsules were touched. The results of the observations are presented in Table 5.

Table 5. Results on the observations of the consistency of the PCM substance in capsules during the experiment.

Time (min)	Temperature (°C)	Comment
0	67.15	Start point. Capsules: hard
2	64.20	Capsules hard
4	61.63	Capsules hard
6	59.65	PCM starts to melt, the capsules are a little bit softer on the edges
8	57.91	PCM is partially liquid; there is a sloshing sound in the capsules from the external side of a tank
10	56.40	The PCM in all capsules is liquid, there is a sloshing sound, but every capsule is still hard in the middle
12	54.73	As above
14	53.59	Four capsules from external sides have fully melted—liquid PCM, there is a sloshing sound, other capsules are still hard in the middle
16	52.32	All capsules are soft throughout their entire volume—PCM is liquid
18	51.16	No changes observed
20	50.36	No changes observed
22	49.59	No changes observed
24	48.92	No changes observed
26	48.32	No changes observed
28	47.83	No changes observed
30	47.42	No changes observed
…	…	…
46	45.19	No changes observed
…	…	…
60	44.16	No changes observed
…	…	…
120	42.15	PCM in all capsules fully liquid

The above experiment allowed us to observe the moment of the phase change of the PCM substance that filled the capsules. Already from the 6th minute (360 s), it was observed that the capsules began to become soft on the edges. This confirms the observation resulting from the previous experiment, indicating that the phase change of the PCM in the capsules occurs quite quickly because of the first minutes of cooling, which cause a much faster drop in the temperature of the water–PCM mixture in relation to the same volume of water alone. During the observations, it was noticed that the capsules located on the outside become soft faster (the PCM melts faster than in the case of capsules located closer to the center of the tank). This is probably due to the way the capsules are arranged, which due to their low density, float close to the water surface, and some of them (those in the middle) are not partially submerged, as can be seen in Figure 9. This is one of the difficulties that may be encountered when trying to apply the tested method of increasing the thermal storage capacity of the heat reservoir by replacing part of the water volume with capsules with the PCM, which is discussed in more detail in Section 4.1. The capsules located in the middle of the tank were still slightly hard in the central part of the capsule in the 14th minute of the experiment, while the capsules located on the outside were already completely soft, and therefore, the PCM contained in them had completely undergone a phase change of melting. All capsules were soft in their entire volume from the 16th minute of observation. They remained so until the 120th minute of the experiment, after which the observation was stopped. At that time, the sound of the splashing liquid PCM could be heard in all capsules.



Figure 9. A photo showing the arrangement of PCM-filled capsules in the tank and how they "float" in the tank.

4. Discussion

4.1. Discussion of the Results and Limitations

The results of the experiment led us to note three characteristic stages of water cooling with PCM-filled capsules. We approximated the course of the temperature change for each with Equations (2)–(4) and also determined the cooling rate. When comparing the average cooling rate of water alone with that of the mixture of water and PCM-filled capsules, it can be seen that it is lower than the cooling rate of the mixture of water and PCM-filled capsules within stages 1 and 2 and higher compared to the slowest final stage 3 of cooling water and PCM-filled capsules. The cooling rates of the mixture of water and PCM-filled capsules for stages 1, 2, and 3 are, respectively, 175×10^{-4} °C/s, 60.8×10^{-4} °C/s, and 9.2×10^{-4} °C/s and the average for water alone at the same time is 19.4×10^{-4} °C/s. These values can be used for the comparison of different PCM-filled capsules in future studies.

During the observations conducted as part of the case 3 experiment, it was observed that the plastic capsules filled with the PCM float on the water. This may cause difficulties in their arrangement in the tank as well as a problem with their mutual blocking when floating below the water surface. During the observations in case 3, it was noticed that the capsules in the middle of the tank will melt the PCM contained in them more slowly. This could be due to the fact that the middle capsules were partially exposed, while the outer capsules had a larger surface area in contact with the water in the tank. The high buoyancy of the capsules may also cause difficulties when trying to load a larger number of them into the tank. For a larger number of capsules, it may be difficult to obtain two or more layers of capsules in the tank because the lower layers will try to float below the surface. The solution may be to use holders that hold the capsules or their group at the desired depth. This is all the more important because in a typical tank there is temperature stratification: in the upper part of the tank, the water has a significantly higher temperature than in the lower part. The method of installation and location in the most advantageous place in the tank will depend on many factors, which in further research on a semitechnical scale should be subject to experimental studies and/or numerical simulations in order to optimize the effect of heat accumulation of the PCM tank.

The choice of encapsulation material for PCMs depends on the specific application needs, including buoyancy control, thermal performance, and durability. The use of a higher density capsule material may be beneficial for several reasons: (i) a heavier capsule will reduce the buoyancy effect; (ii) a denser material may promote increased thermal

conductivity, which in turn may accelerate the heat exchange process between water and the PCM in the capsule; and (iii) a capsule made of such a material may be more durable. On the other hand, the disadvantages may be as follows: (i) the capsule may become harder and less flexible, so space must be provided for the material to expand under the influence of the change of state, (ii) the capsule may be less available and therefore more expensive to purchase, and (iii) a material other than plastic may be susceptible to corrosion. Metals and ceramics provide the best thermal conductivity and mechanical strength but can be expensive and more difficult to fabricate. However, we would like to point out that our research did not include testing the influence of the capsule casing on the heat storage effect. Therefore, we used a commercial product available on the market. Other solutions described in the literature are not available for commercial sale.

It should be noticed that an empty space in the PCM capsules should be provided to accommodate the density variation between the states of matter of the PCM. For now, additional space has not been needed due to the use of a pasty and durable material, but we believe that the choice of the type of capsule material deserves separate research and analysis in this area, such as deciding which one to use, what parameters it should have, and which space for thermal expansion should be used for a given type of capsule. It is most convenient to use a capsule with an appropriate strength that compensates for the volumetric expansion of the PCM by slightly changing its volume.

Other PCMs than those investigated in this experiment may have better cooling performance. Materials with high heat capacity and high thermal conductivity are desirable, which could increase the rate of the cooling process demonstrated in this experiment and, at the same time, increase the difference in the temperature drop (cooling) of water with a PCM compared to water alone. It is worth noting that different PCMs will be adapted to different specific applications of heat storage. This will depend on the purpose of storage (the specificity of heat collection, peak power, the continuity of collection, and thus the characteristics of the tank loading and unloading cycles) but also on the expected temperature of the medium that will be heated using the heat from the storage. In the case of using heat for central heating purposes, the supply and return temperature of the heating water will be important, which is usually variable as a function of the outside air temperature (weather regulation). In the case of using heat to prepare warm utility water, this will be a temperature of approx. 50–55 °C to ensure the appropriate water temperature at the collection point. It should be remembered that cooling performance is not the only criterion for choosing a PCM. The decisive factor in choosing a PCM is the working medium temperature, which should oscillate around the phase change temperature (ideally as the middle of the most common temperature range). In the case of domestic hot water, this is a relatively simple adjustment. Heat storage also occurs using the heat capacity of the solid and liquid after the phase change. These temperature ranges, according to the literature, can be taken approx. ±5 K from the phase change temperature [39–41]. In the paper [3], an analysis of PCM selection based on point values was described. Using this method of PCM selection for domestic hot water purposes, researchers can successfully choose paraffin with 25 carbon atoms with a latent heat of phase change of 238 kJ/kg, thermal conductivity of 0.21 W/(mK), and density of 910 kg/m^3 or salt hydrates $MgSO_4 \cdot 7H_2O$, $Ca(NO_3)_2 \cdot 3H_2O$, and $Zn(NO_3)_2 \cdot H_2O$, characterized by a latent heat of the phase change of 202 kJ/kg, thermal conductivity of 0.163 W/(mK), and density of 490 kg/m^3. However, in our experiment, we focused on the availability of the PCM, choosing one of the few commercial products in the form of capsules. Further considerations on the issues of alternative materials are certainly extremely interesting, which would allow us to expand our research in the future.

This paper does not analyze the costs of producing the material or encapsulating it. This is crucial for practical use in real heating systems. In our opinion, it would be necessary to first develop general guidelines that ensure the most advantageous use of the material adapted to the given type of application and the nature of heat collection. Then, when the design guidelines are known, ensuring the effective use of a PCM in capsules, an economic analysis should be undertaken to answer the question: does the optimized solution bring more benefits than the cost of its production. At the stage of pilot studies on a laboratory scale, this should not determine the reasonableness of the research undertaken because the economic evaluation of an as-yet-imperfect solution will lead to results that, in our opinion, will lower the potential for the implementation of the idea presented by us.

4.2. Perspectives

It is worth noting that there are still many questions, the answers to which can be sought experimentally (by conducting experiments similar to the one presented in this work) or computationally (analytically or numerically). The questions that occurred to us after analyzing the results are as follows.

(1) What is the effect of the arrangement of capsules in the tank on the heat accumulation rate $\Delta t(\tau)$?
(2) What is the optimal distance between the capsules?
(3) Does the shape of the capsules and the amount of the PCM contained in them affect $\Delta t(\tau)$?
(4) Does the amount of water and capsules affect $\Delta t(\tau)$?
(5) How should heat be extracted from the storage tank—from the point of view of its design and possible additional heat exchangers but also the loading and unloading schedule as well as the optimal unloading temperature depending on the purpose of the storage tank?

The work of obtaining an appropriate answer to the above questions is an inspiration for future work on the possibility of using rectangular capsules filled with the PCM RT44HC in existing water energy storage tanks (heat accumulators).

5. Conclusions

The aim of this paper is to compare the cooling rate of the same volume of water as that of the water mixed with the PCM capsules to initially verify the heat storage potential of the capsules. The innovation of this paper is an interesting concept and is the use of plastic capsules filled with the PCM that replace part of the water volume in an existing heat accumulator. The results of the research conducted as part of this laboratory-scale pilot study became the basis for the discussion presented in Section 4, in particular, for proposing five questions for the purposes of future work (see Section 4.2).

In addition, the results and observations from the conducted experimental studies can be summarized with the following conclusions.

- The rate of the decrease in water temperature in the tank with water and PCM capsules during the initial cooling period was 0.010 °C/s and of water alone 0.0026 °C/s (water with the PCM, 40% of the total volume, cools down 3.85 times faster), indicating more intensive heat storage by the water mixture with PCM capsules.
- The decrease in the water temperature in the tank in 2 h was 14 K and 26 K, respectively, in the case of water alone and the water mixture with capsules filled with the PCM, indicating that the PCM stored more heat during this time.
- The cooling process of water with PCM capsules was described by three straight lines given by Equations (2)–(4) in Section 3.1, illustrating the cooling rate in characteristic

periods. They can be used in subsequent studies using other capsule materials and/or other PCMs for comparison to the results obtained for plastic capsules with dimensions of 170 × 85 × 22 mm and a volume of 300 mL filled with PCM RT44HC. A similar characteristic for water alone was approximated by Equation (5).

- The cooling rates of the mixture of water and PCM-filled capsules for stages 1, 2, and 3 are, respectively, 175×10^{-4} °C/s, 60.8×10^{-4} °C/s, and 9.2×10^{-4} °C/s and the average for water alone at the same time is 19.4×10^{-4} °C/s.

The results of the pilot studies provided promising prospects for the use of PCM capsules to increase the heat storage capacity in existing heating system tanks without the need to change their design. Further research on the possibility of using plastic capsules filled with the phase change material in existing heat accumulators should aim to quantify the amount of additionally stored heat and the method of using the capsules on a semi-technical and technical scale, which will be accompanied by challenges related to the operation of the existing heating system.

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Nomenclature

	Symbols
c_p	Specific heat (kJ/(kgK))
g	Thickness of the insulation layer (m)
m	Mass (kg) or (g)
R	Thermal resistance (m^2K/W)
t	Temperature (°C)
V	Volume (dm^3) or (mL)
Δm	Accuracy of mass measurement (single measurement) (g)
ΔV_1	Accuracy of volume measurement volume (single measurement) (mL)
ΔV	Accuracy of volume of capsule measurement (two readings) (mL)
Δt	Temperature difference (°C)
$\Delta \tau$	Period of the time (s)
ΔT_{av}	Accuracy of temperature of water measurement (averaged from four readings) (°C)
Δx	Multiple measurement uncertainty
Δx_i	Uncertainty of single measurement
	Greek symbols
λ	Thermal conductivity (W/(mK))
ρ	Density (kg/m^3)
τ	Time (s)
	Abbreviations
CHP	Combined heat and power (co-generation)
DHW	Domestic Hot Water
PCM	Phase Change Material
PE	Primary Energy

ΔT_1	Accuracy of temperature of water measurement (single measurement) (°C)	PV	Photovoltaic
ΔT_{air}	Accuracy of temperature of air measurement (single measurement) (°C)	RES	Renewable Energy Sources

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